Multi-Branch Retail Performance & Operations Intelligence

Business Analysis Report

Excel + MySQL + SQL + Power BI + DAX

Document Control

Item	Details
Project	Multi-Branch Retail Performance & Operations Intelligence
Geography	Mumbai, Pune, Delhi, Bangalore
Tools	Excel, MySQL 8.0, SQL, Power BI, DAX
Dashboards	5
Validation	Dashboard KPI/chart outputs reconciled against SQL outputs during development

1. Executive Summary

This initiative establishes a comprehensive operational intelligence framework for retail enterprises across four major urban centers - Mumbai, Pune, Delhi, Bangalore.

The objective was to transform raw retail data into an interactive business intelligence solution that enables management to monitor sales, store performance, products, customers, shipments, returns, refunds, and operational performance.

Initial data analysis and profiling was conducted in Excel, followed by exploratory investigations, rigorous structured analysis and validation using MySQL. Power BI and DAX served as the primary environment for sophisticated data modeling and the formulation of critical KPI development and interactive reporting.

The final solution contains five dashboards: Executive Overview; Store Performance; Product & Category Performance; Customer Performance; and Operations, Returns & Shipments.

This analysis identifies strong sales growth, Mumbai's leading commercial position, strong repeat-customer behavior, product/category opportunities, and important operational attention areas around late shipments and returns. Major Power BI KPIs and analytical outputs were reconciled against SQL before finalization.

The result is a decision-support solution that converts retail data into validated KPIs, interactive dashboards, business insights and management recommendations while explicitly documenting the limits of the dataset.

1.1 Business Problem

The retail organization operates across multiple cities and generates large volumes of transactional, customer, product, payment, shipment, and return data. Management needs a centralized view of commercial and operational performance.

- Identify strongest and weakest cities/stores.
- Measure sales, orders, AOV and growth.
- Identify high-performing products and categories.
- Understand customer purchase frequency, value and repeat behavior.
- Monitor shipment status and late delivery.
- Measure returns and refund value.
- Identify revenue, customer-value and operational improvement opportunities.

1.2 Business Task

Analyze the retail data of the four-city business to provide management with a comprehensive, validated view of sales, store, product, customer and operational performance.

- Determine which cities/stores perform best across sales, orders, customers and AOV.
- Identify top products and categories and understand their sales contribution.
- Measure customer size, purchase frequency, customer value and repeat-customer performance.
- Evaluate shipment volumes, status distribution and late-delivery performance.
- Compare return rates across cities and quantify returned items and refund value.
- Identify business opportunities and operational priorities supported by the available data.
- Provide management-ready Power BI dashboards whose critical outputs are independently validated against SQL.

1.3 Objective

- Analyze overall retail sales performance and identify key revenue-driving markets.
- Compare performance across Mumbai, Pune, Delhi, and Bangalore to identify high and low performing cities.
- Identify top performing products and product categories and understand their contribution to overall sales.
- Analyze customer purchasing behaviour, customer value, purchase frequency, and repeat-customer performance.
- Evaluate shipment performance and identify patterns in late deliveries across cities.
- Analyze returns and refunds to identify potential operational and revenue-impact areas.
- Develop interactive Power BI dashboards that provide management with a clear view of sales, customers, products, and operations.
- Translate the analysis into actionable business insights and recommendations for improving revenue, customer value, and operational efficiency.
- Document data limitations so conclusions do not exceed what the data supports.

1.4 Dataset

The analysis uses a 12-table retail data warehouse dataset containing approximately 1.59 million records across the warehouse tables.

Table	Number of Records
Categories	30
Customers	50,000
Employee	1000
Order Items	600,000
Orders	300,000
Payments	300,000
Products	10,000
Promotions	50
Returns	30,000
Shipments	300,000
Stores	100
Suppliers	200
Total	**1591380**

1.5 Technology Stack

- Excel
- MySQL
- Power BI
- Google Docs
- Google Slides
- GitHub

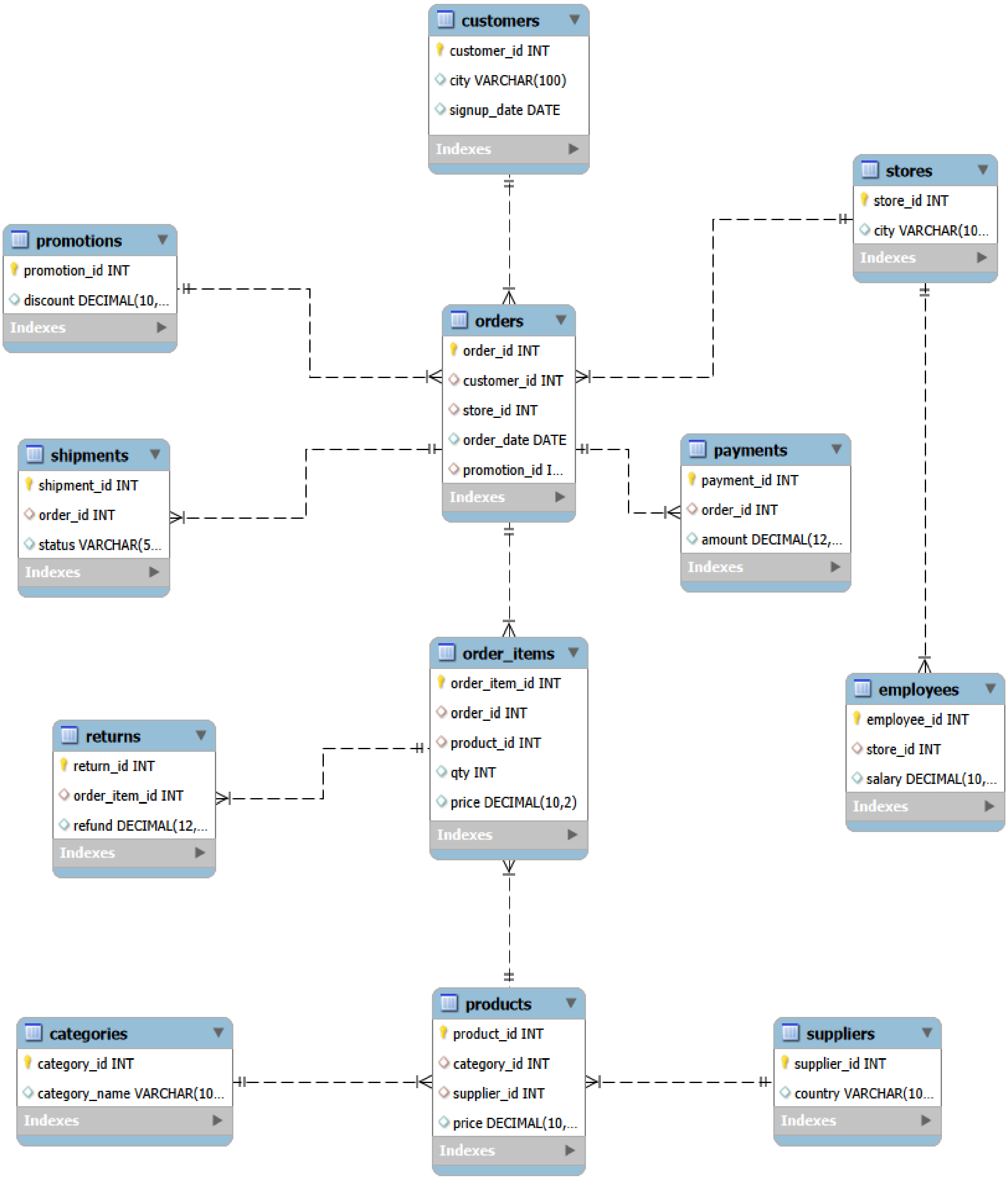
Tools	Purpose
Excel	Initial data exploration
MySQL 8.0	Data warehouse querying, EDA, aggregation and validation
SQL	EDA, joins, aggregations, distinct counts, KPI calculations and validation
Power BI	Data modeling, relationships, slicers and dashboard
DAX	Reusable measures, ratios and calculated columns
Google Docs	Project Documentation
Google Slides	Project presentation
GitHub	Portfolio hosting and version control

1.6 Analytics Workflow

2. Data Model

Core analytical flow:

- Customers → Orders → Order Items → Returns
- Orders → Shipments



Relationship and filter propagation were tested during development, particularly for city-level shipment and return analysis.

3. Business Insights

3.1 Dashboard 1: Executive Overview

<u>Key Findings</u>

- Gross sales are approximately 3.83B.
- YoY sales growth is 29.77%.
- AOV is approximately 12.76K.
- Mumbai is the strongest revenue market.
- Q4 is the strongest quarter.

<u>Management Conclusion</u>

The retail business generated approximately ₹3.83B in gross sales across four cities, with Mumbai contributing the highest sales. Sales grew 29.77% year over year, while average order value reached ₹12.76K. Revenue remained relatively stable across months, with Q4 emerging as the strongest quarter.

3.2 Dashboard 2: Store Performance

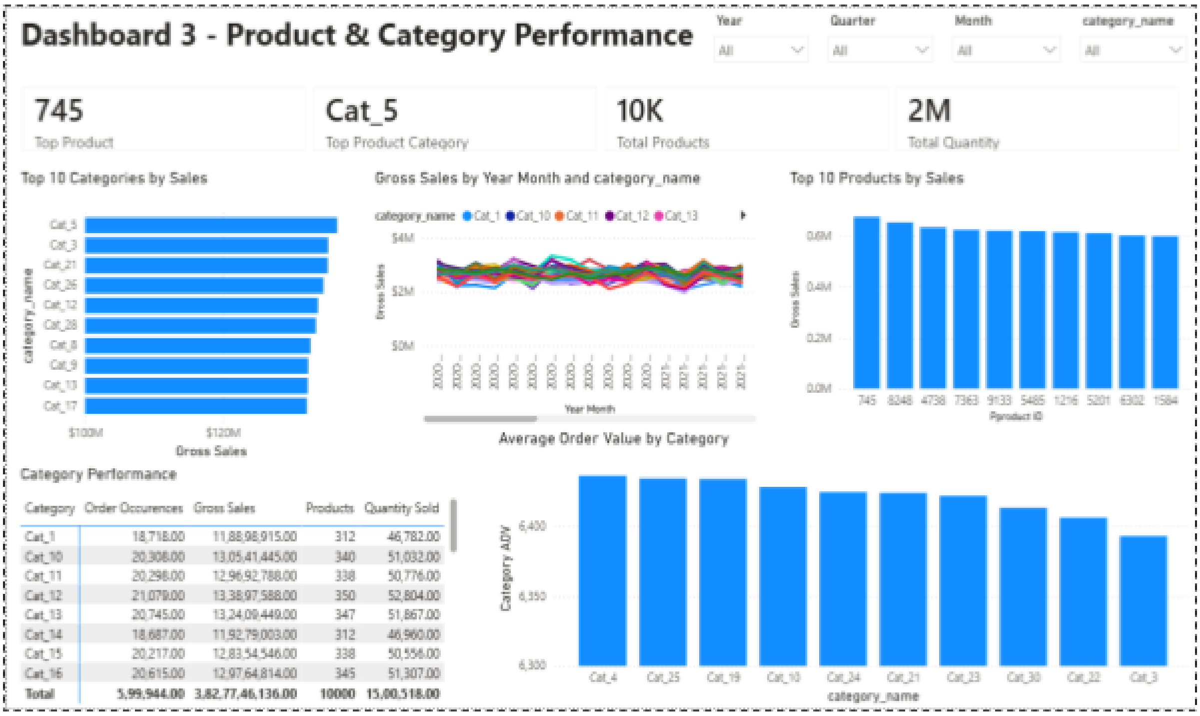
<u>Key Findings</u>

- Mumbai is the strongest-performing market.
- Mumbai leads order volume and customer base.
- Mumbai has the strongest AOV position in the store comparison.
- Mumbai contributes approximately 31% of overall sales.
- Category performance varies by city, supporting localized strategy.

<u>Management Conclusion</u>

Mumbai is the strongest-performing branch, leading in orders, customers, average order value and overall sales contribution. Pune follows Mumbai, while Delhi and Bangalore show comparatively lower performance. Mumbai contributes approximately 31% of total sales, indicating a significant concentration of revenue in one market. Category-level performance also varies between cities, highlighting the need for location-specific product and inventory strategies.

3.3 Dashboard 3: Product & Category Performance



Key Findings

- Product 745 is the top product.
- Cat_5 is the top category.
- Approximately 10K distinct products are represented.
- Approximately 2M units are represented.
- Leading categories are relatively diversified.
- Category AOV varies and creates mix/cross-selling opportunities.

Management Conclusion

Product and category performance is relatively diversified, with Cat_5 emerging as the leading category and Product 745 as the top-selling product. The company manages approximately 10K products and has generated around 2M units in quantity sold. While leading categories generate similar levels of revenue, differences in category AOV indicate opportunities to optimize product mix, pricing and cross-selling. Monitoring category trends over time can further improve inventory and promotional planning.

3.4 Dashboard 4: Customer Performance

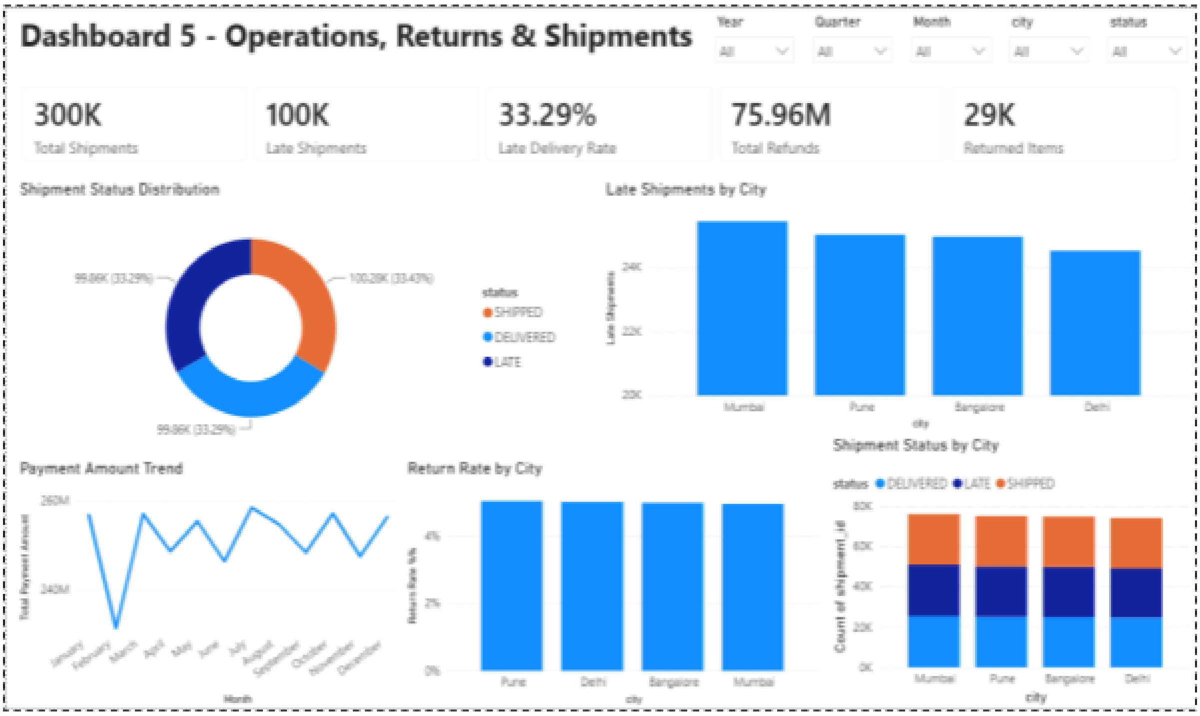
Key Findings

- Mumbai has the largest customer base and highest customer sales.
- Average orders per customer are around 6.
- Repeat Customer Rate is 98.56%.
- The largest customer-value segment is 50K–100K.
- Customer 33455 is the top customer.
- High-value customers are a smaller segment.

Management Conclusion

Customer performance is strong across all four cities, with Mumbai leading in customer count and customer sales. Purchase frequency is relatively consistent, averaging approximately six orders per customer, indicating that revenue differences are primarily associated with customer base size and customer value. The 98.56% repeat customer rate indicates strong customer retention, while the customer value distribution shows that the largest segment falls within the 50K–100K range. Increasing the proportion of customers moving into higher-value segments presents a significant opportunity for revenue growth.

3.5 Dashboard 5: Operations, Returns & Shipments Performance



Key Findings

- 99,855 of 300,000 shipments are late, producing a 33.29% late-delivery rate.
- Mumbai has the highest absolute late-shipment count; absolute count alone should not be called the worst rate.
- Shipment volume is almost evenly distributed by status.
- Pune has the highest return rate at 5.04%.
- Refunds total 75.96M.
- 29,251 order items were returned.

Management Conclusion

Operations analysis reveals a significant delivery challenge, with 99,855 of 300,000 shipments classified as late, resulting in a 33.29% late-delivery rate. Mumbai records the highest absolute number of late shipments, while Pune has the highest return rate at 5.04%. The business also processed approximately ₹75.96M in refunds and 29,251 returned order items. These results indicate that delivery reliability and return reduction are two important opportunities for improving operational efficiency and protecting revenue.

4. Consolidated Validated KPI Register

Dashboard/Area	Metric	Validated value
D1 Executive	Gross Sales	~3.83B
D1 Executive	YoY Sales Growth	29.77%
D1 Executive	AOV	12.76K
D2 Store	Top Market	Mumbai
D2 Store	Mumbai Sales Contribution	~31%
D3 Product	Top Product	745
D3 Product	Top Category	Cat_5
D3 Product	Distinct Products	~10K
D3 Product	Total Quantity	~2M
D4 Customer	Total Customers	~50K
D4 Customer	Active Customers	49,886
D4 Customer	Repeat Customers	49,168
D4 Customer	One-Time Customers	718
D4 Customer	Repeat Rate	98.56%
D4 Customer	Avg Orders/Customer	6.01
D4 Customer	Avg Customer Value	76.73K
D4 Customer	Top Customer	33455
D5 Operations	Total Shipments	300,000
D5 Operations	Late Shipments	99,855
D5 Operations	Late Delivery Rate	33.29%
D5 Operations	Refunds	75,962,300
D5 Operations	Returned Items	29,251

5. Final Business Story

The four-city retail business generated approximately ₹3.83B in gross sales with 29.77% year-over-year growth. Mumbai is the strongest market across sales, orders and customer value, while customer retention is exceptionally strong at 98.56%. The largest growth opportunity is therefore not simply customer acquisition, but increasing the value of existing customers. Product performance is diversified across approximately 10K products, with Cat_5 and Product 745 leading their respective rankings. Operationally, the most significant concern is delivery reliability, with 33.29% of shipments classified as late, alongside ₹75.96M in refunds and 29,251 returned items. Management should prioritize delivery improvement, reduce avoidable returns, protect high-performing products and markets, and use customer-value strategies to drive further revenue growth.

6. Business Recommendation

Recommendation 1: Improve Delivery Reliability

Reduce the current 33.29% late-delivery rate by investigating shipment processing delays, city-level operational bottlenecks, and monthly delivery patterns. Establish monthly delivery targets and prioritize initiatives that can improve overall shipment reliability.

Recommendation 2: Increase Customer Value

Leverage the 98.56% repeat customer rate by focusing on existing customers through loyalty programs, personalized promotions, cross-selling, upselling, and product bundles. Increasing the value of existing customers can support sustainable revenue growth.

Recommendation 3: Reduce Avoidable Returns

Investigate return reasons, high-return products, high-return categories, and city-level return patterns to identify preventable returns. Prioritize high-value returned products and monitor refund trends alongside return volumes to reduce unnecessary costs and protect refund value.

Recommendation 4: Strengthen Market Performance

Maintain appropriate inventory and operational support for Mumbai, the strongest revenue market, while analyzing the factors contributing to its performance. Replicate successful product, category, and customer strategies in Pune, Delhi, and Bangalore where appropriate to improve overall market performance.

Recommendation 5: Optimize the Product Portfolio

With approximately 10K products, identify high-performing, low-performing, high-AOV, and high-return products to support better inventory allocation and SKU rationalization. Continue protecting strong performers such as Product 745 and Cat_5, while reviewing weaker products for optimization opportunities.

Recommendation 6: Leverage Q4 Performance

Q4 is the strongest quarter by gross sales, indicating an opportunity to use the strategies and demand patterns observed during this period to strengthen performance throughout the year. Analyze the products, categories, cities, and customer segments contributing to Q4's stronger performance and use these insights to plan inventory, promotions, and marketing activities ahead of weaker quarters.

7. Data Limitations & Analytical Caveats

7.1 Source-Data Limitations

- **No explicit product cost/COGS:**
  - Selling-price information is available but explicit product cost/COGS is not.
  - True gross profit, gross margin, net profit and accounting profitability therefore cannot be derived reliably.
  - Revenue/transaction value must never be relabeled as profit.
- **Customer signup-date inconsistency:**
  - 120,020 orders have order_date earlier than the associated signup_date.
  - Signup_date is unreliable for tenure, acquisition cohorts and pre/post-signup behavior.
  - Orders should be retained and signup_date excluded from those analyses.
- **No return-date field:**
  - True return-month trends, return timing and time-to-return cannot be calculated.
  - Time grouping of returns must be described as association with the original order month/date.
- **Multiple return records can reference one order item**:
  - Counting return records as unique returned items can inflate return rates.
  - Use COUNT(DISTINCT order_item_id) and appropriate refund aggregation.
- **Incomplete/partial final time period:**
  - The final period may not be comparable with complete earlier periods.
  - Apparent end-of-series declines can be exaggerated.
  - Use like-for-like comparisons.
- **No payment method/status:**
  - The payments data available to the project does not contain payment method or payment status, so those analyses are not claimed.
- **Shipment status is limited to SHIPPED, DELIVERED and LATE:**
  - detailed carrier, route, warehouse and SLA attributes are unavailable.
- The dataset is historical/static rather than a live operational feed.

7.2 Analytical Caveats & Interpretation Rules

- Use order_items.qty × order_items.price for historical sales calculations; products.price is the master/product price while order_items.price is the transaction-level selling price.
- Promotion discount application is not fully specified. Do not subtract promotions.discount again unless the transaction-price semantics are confirmed, because this could double-count the discount.
- Promotion comparisons are associational, not causal. Use 'associated with' rather than 'caused by'.
- Payment-to-order-value differences require investigation and are not automatically payment errors.
- Gross transaction value is not profit; use gross sales, transaction value or revenue.
- Refunds must be aggregated at the correct order-item grain because an item can have multiple return/refund records.
- Return rate requires a defined denominator; for order-item return rate use distinct returned order_item_id divided by total order-item records.
- Shipment/return relationships are descriptive. A higher return rate among shipment statuses does not prove shipment status caused the return.
- Order-date-based return trends are associations with the original order period, not actual return timing.
- Absolute late-shipment counts should not alone define the worst city; normalized rates are more appropriate for fair comparison.

7.3 Supported vs Unsupported Claims

- **Supported metrics:** gross sales/transaction value, order volume, units sold, AOV, purchase frequency, store/product/category/supplier sales distributions, descriptive promotion comparisons, payment distributions and reconciliation, shipment-status distributions, distinct order-item return rates and aggregated refunds.
- **Not supported without additional data:** gross profit, net profit, profit margin or true profitability.
- **Not supported:** reliable customer tenure or acquisition cohorts based on signup_date, actual return-date trends or time-to-return, causal claims that promotions increase sales, causal claims that shipment status causes returns, subtracting promotions.discount again to create post-discount net sales without confirmed price semantics.

7.4 Preferred Terminology

Use	Avoid
Gross sales / transaction value / revenue	Profit / gross profit / profit margin
Promotion-associated sales	Sales caused by promotion
Distinct returned order items	Return records treated as unique returned items
Returned items associated with original order month	Actual monthly returns
Payment difference / reconciliation difference	Payment error
Higher return rate among late-shipment orders	Late shipment caused the return

8. Conclusion

The Multi-Branch Retail Performance & Operations Intelligence Project successfully transforms retail data into a structured, validated business intelligence solution covering sales, store performance, products and categories, customers, shipments, returns, and refunds across Mumbai, Pune, Delhi, and Bangalore.

Using Excel for initial data inspection, MySQL/SQL for detailed analysis and independent validation, and Power BI with DAX for data modeling and dashboard development, the project delivers five interactive dashboards that provide a comprehensive view of business and operational performance.

The analysis highlights 29.77% year-over-year sales growth, Mumbai's strong commercial performance, a 98.56% repeat customer rate, and a 33.29% late-delivery rate, while also identifying significant refund and return activity. These findings provide clear opportunities to improve delivery reliability, increase customer value, reduce avoidable returns, optimize the product portfolio, strengthen market performance, and leverage stronger quarterly performance.

All major Power BI KPIs and analytical outputs were validated against SQL results, improving confidence in the reported metrics. At the same time, documented data limitations ensure that the conclusions remain within what the dataset can reliably support, particularly for profitability, return timing, customer tenure, and casual analysis.

Overall, the project demonstrates a complete Data Analyst / Business Intelligence workflow from data inspection and SQL analysis to data modeling, DAX, dashboard development, validation, business insights, and recommendations, providing a practical framework for data-driven retail decision-making.

Appendix

- **Late Shipments**: SQL 99,855; Power BI displayed 100K due to display-unit rounding; underlying value matched.
- **Total Refunds**: SQL 75,962,300.00; Power BI displayed 75.96M; underlying value matched.
- **Returned Items:** SQL 29,251; Power BI displayed 29K; underlying value matched.
- Customer Value Distribution uses an independent numeric sort column.
- City-level shipment and return visuals required correct relationship/filter propagation.
- Dashboard 3 KPI was renamed from Total Products Sold to Total Products because it counts distinct product IDs.
- Payment method/status analysis is not claimed because those fields are unavailable.
- The supplied limitations register is the master limitations register for the project and should be updated only when later SQL analysis verifies or clarifies a limitation.

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